Yanhua Lu, Ph.D.

Assistant Professor, Extension Specialist
Department of Landscape Architecture
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Research Interests

Research exploring how design can respond to people's experiences, community knowledge, and changing environmental conditions. Areas of interest include community participation, environmental perception and behavior, resilience, and design education. Applying participatory, mixed-methods, geospatial, and immersive approaches to support evidence-based design and planning.

Education

2019 - 2024	**Ph.D. in Design** North Carolina State University, Raleigh, NC, United States Dissertation: *Exploring the Effects of Augmented Reality (AR) Wayfinding Tools on Perceived Urban Walkability and Pedestrian Behavior in the Digital Era* Advisor: Dr. Celen Pasalar
2019 - 2021	**Graduate Certificate in Geographic Information Systems (GIS)** Center for Geospatial Analytics, North Carolina State University
2014 - 2016	**Master of Landscape Architecture** University of Massachusetts Amherst, Amherst, MA, United States Thesis: *Creating a Successful Wayfinding System: Lessons Learned from Springfield, Massachusetts* Advisor: Prof. Michael DiPasquale
2010 - 2014	**Bachelor of Engineering in Landscape Architecture** Beijing Forestry University, Beijing, China

Academic Appointments

	Assistant Professor, Extension Specialist
2025 - present	Department of Landscape Architecture, College of Design (CoD), Iowa State University (ISU) Community and Economic Development, ISU Extension and Outreach
	Postdoctoral Research Scholar
2024 - 2025	Under the supervision of Dr. Gavin Smith and Dr. Celen Pasalar, Department of Landscape Architecture and Environmental Planning, North Carolina State University (NC State)
	Graduate Research Assistant
2023 - 2024	FEMA-funded *University DTA Network Project*, NC State

2021 - 2023	NCDOT-funded *Urban Applications of Innovative Intersections Project*, in partnership with the Institute for Transportation Research and Education (ITRE), NC State
2019 - 2021	PhD in Design Program, College of Design, NC State
2015 - 2019	UMass Design Center, Department of Landscape Architecture and Regional Planning, UMass Amherst
	Graduate Teaching Assistant
2020 - 2023	Department of Landscape Architecture and Environmental Planning, NC State
2016	Department of Landscape Architecture and Regional Planning, UMass Amherst

Scholarship

Books

2026 Sanoff, H., & **Lu, Y.** (2026). *Historical Roots of EDRA: Environmental Design Research Association*.

Peer-Reviewed Conference Proceedings

2026 **Lu, Y.,** & Pasalar, C. (2026). Navigating with awareness: How wayfinding tools alter perception, safety, and urban experience. *Proceedings of the International Association of People-Environment Studies (IAPS) Conference,* University of Surrey, Guildford, United Kingdom.

Lu, Y., & Pasalar, C. (2026). Perceiving Safety and Environmental Awareness through AR Pedestrian Navigation: A Comparative Study. *Proceedings of the 57th Environmental Design Research Association (EDRA 57) Conference,* Amherst, MA.

Pasalar, C., & **Lu, Y.** (2026). Empowering Communities through Co-Created Design: Pedagogy and Impact. *Proceedings of the 57th Environmental Design Research Association (EDRA 57) Conference,* Amherst, MA.

Peery, J., Pasalar, C., & **Lu, Y.** (2026). Geogames for Community Engagement: Learning and Experimentation for Climate-Responsive Planning and Community Health. *Proceedings of the 3rd Geogames Symposium,* Ames, IA.
https://doi.org/10.31274/geogames.25880

2025 **Lu, Y.** (2025). Designing Resilience: Community-Driven Green Infrastructure and Sustainable Land Use Planning. *Proceedings of the Urban-Rural Systems (URSYS) Research Network Conference,* Davenport, IA.
https://arcg.is/0Pravn2

Peer-Reviewed Conference Presentations

2026 **Lu, Y.,** & Pasalar, C. (2026, June 29 - July 3). *Navigating with awareness: How wayfinding tools alter perception, safety, and urban experience.* International Association of People-Environment Studies (IAPS) Conference, University of Surrey, Guildford, United Kingdom.

Lu, Y., & Olson-Douglas, E. (2026, June 15-18). *An Extension-Anchored Partnership for Addressing Aging Housing and Neighborhood Revitalization through Design Education.* National Association of Community Development Extension Professionals (NACDEP) Annual Conference, Wichita, KS.

Lu, Y., & Pasalar, C. (2026, May 27-30). *Perceiving Safety and Environmental Awareness through AR Pedestrian Navigation: A Comparative Study.* 57th Environmental Design Research Association Annual Conference (EDRA 57), Amherst, MA.

Pasalar, C., & **Lu, Y.** (2026, May 27-30). *Empowering Communities through Co-Created Design: Pedagogy and Impact.* EDRA 57, Amherst, MA.

Peery, J., Pasalar, C., & **Lu, Y.** (2026, May 3-6). *Geogames for Community Engagement: Learning and Experimentation for Climate-Responsive Planning and Community Health.* 3rd Geogames Symposium, Ames, IA.

2025 **Lu, Y.** (2025, October 3-5). *Designing Resilience: Community-Driven Green Infrastructure and Sustainable Land Use Planning.* Urban-Rural Systems (URSYS) Research Network, Davenport, IA.

Invited Lectures and Presentations

2026 **Lu, Y.** (2026, October 27). *Smart Mobility, Health, and Well-being.* LAR 582: Smart and Healthy Cities, NC State, Raleigh, NC.

Lu, Y. (2026, March 30). *Designing Together: Shaping Places with Communities.* Institute for Design Research and Outreach (IDRO) Research Seminar, ISU, Ames, IA.

Lu, Y. (2026, March 3). *Walkability Perception and Assessment.* CRP 4320: Community Planning Studio, ISU, Ames, IA.

Lu, Y. (2026, February 17). *Community-Based Design Approach.* LA 5430: Community Planning Studio, ISU, Ames, IA.

2025 **Lu, Y.** (2025, November 5). *Advancing Community Awareness of Hazard Mitigation and Preparedness.* Priority Program Initiative Convening, ISU, Ames, IA.

Lu, Y. (2025, October 1). *Community Participation Techniques.* LA 3010: Site Design II Studio, ISU, Ames, IA.

Lu, Y. (2025, October 1). *Community Participation Techniques in Design & Planning.* CRP 4320: Community Planning Studio, ISU, Ames, IA.

Lu, Y. (2025, September 5). *Community-Based Design Research.* LA 4040: Advanced Landscape Architectural Design Studio, ISU, Ames, IA.

Teaching

Courses Taught

Fall 2026	**LA 4780M/5780M,** Topical Studies in Landscape Architecture: Social/Behavioral *Community Participation Techniques in Design and Planning* 3 credits; offered to advanced-level undergraduate and graduate students in CoD Enrollment: 5 students, including 4 undergraduate students (1 fifth-year, 2 fourth-year, and 1 returning third-year) and 1 graduate student, all in Landscape Architecture
Spring 2026	**DSNS 5460,** Interdisciplinary Design Studio *HOMEGROWN: Reframing the Future of Small-Town Communities and Housing* Co-instructor with Anna Aversing 6 credits; offered to advanced-level undergraduate and graduate students in CoD Enrollment: 19 students, including 18 undergraduate students (8 fifth-year Architecture, 6 fifth-year Landscape Architecture, 3 fourth-year Interior Design, and 1 fourth-year Community and Regional Planning) and 1 graduate student in Interior Design

Undergraduate Research Mentoring

2026 - present	**Jack McMenamy,** Architecture, Class of 2026 Research assistant for the Fairfield HOMEGROWN project; contributed to the development and production of the studio publication and post-studio project deliverables.
2026, April - September	**Aidan Naditz,** Landscape Architecture, Class of 2027 Research assistant for the Coralville stormwater research and Extension project; contributed to community participation activities, educational materials, visualization, field-day programming, and project deliverables.
2026, March - August	**Jadan Tseng,** Industrial Design, Class of 2027 Research assistant for the Bio-inspired Speculative Design project; contributed to the qualitative data transcription, preliminary coding, and content analysis.

Graduate Thesis Advising

2026 - present	**Mouri Tabassum,** Master of Landscape Architecture, anticipated 2027 Working title: *Exploring the Influence of Urban Environmental Features of residential Areas on Perceived safety versus Actual Safety Using Augmented Reality* Major Professor. Thesis research examining women's perceived safety in relation to urban environmental characteristics. Advising development of a research poster for submission to the 2027 CELA Annual Conference.

Graduate Research Mentoring

2026 - present	**Rocio Melendez,** Master of Industrial Design, anticipated 2027

	Research assistant for the Bio-inspired Speculative Design project; conducted qualitative content analysis and initiated development of an educational game prototype.
2026, April - September	**Alex Boateng,** Master of Environmental Science, anticipated 2027 Research assistant for the Coralville stormwater project; contributed to Field Day program development and community participation activities.

Curricular Development

2026 Fall	*Community Participation Techniques in Design and Planning.* Developed and taught a new departmental elective, LA 4780M/5780M, including the course structure, content, readings, assignments, and participatory learning activities. Initiated development of the course as an X-course to support future interdisciplinary enrollment across the College of Design.
2026 Spring	*HOMEGROWN: Reframing the Future of Small-Town Communities and Housing.* Co-developed and co-taught an interdisciplinary, community-based design studio with Anna Aversing. Developed the studio in partnership with the City of Fairfield and local partners as a pilot model for connecting interdisciplinary design education, Extension, and community-based projects.

Student Work Exhibitions and Showcases

2026 Spring	*HOMEGROWN: Reframing the Future of Small-Town Communities and Housing.* Student final work exhibited at the IGNITE Innovation Showcase, May 5-8, Student Innovation Center, Iowa State University. https://sictr.iastate.edu/innovation-programs/ignite-showcase/ignite-week-long-displays/

Guest Reviewer

2026	LA 6020: Designing the Living Environment, ISU
	LA 3020: Ecological Design, ISU
	DSNS 5460: Interdisciplinary Design Studio, ISU
	ARTID 6680: Advanced Experimental Interior Design, ISU
	LAR 582: Smart and Healthy Cities, NC State
	LAR 540: Research Methods, NC State
2025	LA 6010: Introduction to Landscape Architecture, ISU
	LA 4040: Advanced Landscape Architectural Design, ISU
	LA 4020: Urban Design, ISU
	LA 3010: Site Design II, ISU
	DSNS 5010: Introduction to Research Design, ISU
	DSNS 1020: Design Studio I, ISU
	ARCH 3010: Architectural Design III, ISU
	LAR 582: Smart and Healthy Cities, NC State
	ARC 503/LAR 507: City Design Studio, NC State

Grants for Teaching

2025 – 2026 *HOMEGROWN: Community-Driven Housing and Neighborhood Revitalization in Fairfield, Iowa*
Principal Investigator (PI). Curriculum-based Design Outreach Grant, IDRO. $3,000.

Research

Completed Projects and Programs

2026
April - September *Coralville as a Model for Clean Stormwater Resiliency*
Interdisciplinary research and Extension project examining stormwater practices, public education, and innovative engineered materials in collaboration with the City of Coralville. Led design visualization, community participation, educational outreach, evaluation, and synthesis components.

Current Projects and Programs

2026 – present *Student Learning Outcomes in Community-Based Design Education*
Examining how community-based design education influences student learning, community participation practices, interdisciplinary collaboration, and design decision-making through pre- and post-studio evaluation.

2026 – present *A Playful Ecological Awareness: Bio-inspired Speculative Design as a Design-Based Learning Case*
Examining how ecology-themed play, bio-inspired design, speculative storytelling, and reflection support ecological awareness among undergraduate students. Data collection completed; qualitative analysis and educational-tool prototyping underway.

2025 – present *Historical Roots and Continuing Influence of EDRA*
Collaborative research with Henry Sanoff examining the historical development and continuing influence of EDRA through interviews with scholars in environmental design research. The project builds on the completed book *Historical Roots of EDRA: Environmental Design Research Association.*

2025 – present *How Augmented Reality Wayfinding Influences Walkability, Environmental Perception, and Travel Behavior*
Extending dissertation research through journal manuscript development examining relationships among AR wayfinding, perceived walkability, environmental perception, safety, and pedestrian travel behavior.

Future Projects and Programs

Planned How Smartglasses Influence Perceived Safety, Travel Behavior, and Environmental Perception

Planned How Alternative Intersection Design Influences Community Development, Well-being, and Sense of Belonging

Grants for Research / Creative Activities

2026 *Coralville as a Model for Clean Stormwater Resiliency*
Co-PI; PI: Dr. Joe Charbonnet. Adaptive and Sustainable Natural and Built Environment Summer Fellowship, ISU Extension and Outreach. $25,000 total; $8,174 allocated to Lu.

2025 – 2026 *A Playful Ecological Awareness: Bio-inspired Speculative Design as a Design-Based Learning Case*
Co-PI; PI: Dr. Miray Boga. Signature Theme Networking Grant, ISU IDRO. $7,500 total; $2,500 allocated to Lu.

Service Related to Research / Creative Activities

2026 - 2027 Guest Editor, Special Issue: *Computational Approaches to Landscape and Urban Systems: Analysis, Simulation, and Design Practice,* Land

Abstract Reviewer

2026 Council of Educators in Landscape Architecture (CELA) Annual Conference

2026 Environmental Design Research Association (EDRA) Annual Conference

2026 Association of Collegiate Schools of Architecture (ACSA) Annual Conference

2026 National Association of Community Development Extension Professionals (NACDEP) Annual Conference

2026 3rd Geogames Symposium

Manuscript Reviewer

2026 Environment and Planning B: Urban Analytics and City Science

2026 Landscape Research Record, Volume 15

Extension and Outreach

Reports/Plans/Designs/Other Deliverables

2026 Coralville as a Model for Clean Stormwater Resiliency:
https://sites.google.com/iastate.edu/isu-coralvillesummerfellowship
Developed a public-facing project website documenting stormwater research, educational resources, project activities, and outcomes. [Project Website]

2026 Coralville Stormwater Educational Signage
Developed public-facing educational signage communicating stormwater practices, infrastructure, and project research. [Signage Design]

2026 **Lu, Y.,** McMenamy, J., Naditz, A., & Aversing, A. (2026). *HOMEGROWN: Reframing the Future of Small-Town Communities and Housing through Community-Based Design.* Ames, IA: Iowa State University. [Project Report]

Projects

2026 - present *An Ecosystem for Community Design: Connecting the Iowa Mayors' Design Workshop, Design Education, Extension, and Community Partnerships*
Developing an integrated framework that connects Extension, community-based design education, and applied research to support sustained partnerships with

	Iowa communities. Builds on the Iowa Mayors' Design Workshop and community-based studios, with Cherokee and Fairfield serving as initial community contexts for developing and refining the framework.
2026 – present	*Fairfield Housing Pattern Book* Developing an implementation-oriented housing rehabilitation pattern book that translates HOMEGROWN studio research and design recommendations into an accessible resource for Fairfield partners.
2026 - present	*Iowa Mayors' Design Workshop (IMDW) Evaluation Program* Developing an evaluation framework and standard operating procedure to assess the educational, collaborative, and community impacts of the Iowa Mayors' Design Workshop across participating mayors, communities, and ISU partners.

Educational Outreach

2026 September 4	*Coralville Stormwater Research Field Day* Coralville Public Library, IA Co-led a statewide field day for water professionals integrating site-based learning, research presentations, facilitated discussions, and program evaluation. 42 participants representing water professionals, engineers, designers, planners, Extension specialists, municipal staff, and legislators.
2026 August 8	*Community Event at Coralville 5th Street Social* 5th St, Coralville, IA Developed and facilitated interactive stormwater education activities for the public.
2026 March 6	*HOMEGROWN Community Workshop* Jefferson County Extension Office, Fairfield, IA Developed and facilitated participatory activities with students and community partners to gather community insights related to housing and neighborhood revitalization.

Grants, Contracts and Fees for Extension / Professional Practice / Engagement

2026 - 2027	*An Ecosystem for Community Design: Connecting the Iowa Mayors' Design Workshop, Design Education, Extension, and Community Partnerships* I-Fellow. Innovation and Entrepreneurship (I+E) Fellowship, ISU CoD. $5,000.

Service Related to Extension / Professional Practice / Engagement

2025 – 2026	Priority Program Initiative (PPI) Fellowship Review Committee, ISU Extension and Outreach
2025 – present	Member, Iowa Extension Disaster Education Network (EDEN)
2025 - present	Member, ISU Extension and Outreach Priority Program Initiative (PPI) Core Team

Media Coverage

2026	Coverage of the HOMEGROWN Community Workshop, Fairfield, Iowa: Hallman, A. (2026, March 11). *Workshops gather public input on Fairfield's housing, amenities*. Southeast Iowa Union.

https://www.southeastiowaunion.com/workshops-gather-public-input-on-fairfield-s-housing-amenities/article_6344e446-3b71-4a65-8de0-a0fdca2dc7e3.html

Professional Development

2026 - 2027	Research Collaboration Catalyst, selected cohort member, ISU
2026	Teaching Partners Program, Center for Excellence in Learning and Teaching (CELT), ISU
2026	Faculty Success Program, National Center for Faculty Development and Diversity (NCFDD)
2025 - present	Game Studies Learning Community, ISU
2025	Grant Writer's Seminars and Workshops, Office of the Vice President for Research (OVPR), ISU
2025	Advanced Teaching Certificate, NC State
2025	Communication Skills Certificate, NC State
2024	Teaching Certificate, NC State

Professional Affiliations

2019 - present	Member, Environmental Design Research Association (EDRA)
2019 - present	Member, Council of Educators in Landscape Architecture (CELA)
2014 - present	Member, American Society of Landscape Architects (ASLA)